June

Week 1: (06/22-06/26) **Present: Hannah Bae, Ella Yu, Toni Densmore, Kavi Shrivastava**

Project Date	06/22	06/23	06/24	06/25	06/26
Experiments /Procedures	Alphafold practice Village project Incubating LB Broth	Village project Practice protocols (ELISA, Transformation)	-	-	-
Key Results	-	-	-	-	-

July

Week 2: (06/29-07/03) **Present: Hannah Bae, Ella Yu, Toni Densmore, Kavi Shrivastava**

Project Date	07/01	07/02	07/03
Experiments /Procedures	Outline Protocols: HIS-TAG, IPTG SPIKING, SULFOTRANSFERASE ASSAY, SDS-PAGE	Outline protocols and get them approved Prepared for SDS-PAGE (Destaining Solution, Methanol 200mL, Deionized Water 160 mL, Acetic Acid 40 mL, Staining solutions– Brilliant Blue R-250 10 mL's| 90 mL's Nanopure Water)	Researching the distribution Kit (Fluorescent receptor, Sumo)
Key Results	-	-	-

Week 3: (07/06-07/10) **Present: Hannah Bae, Ella Yu, Toni Densmore, Kavi Shrivastava**

Project Date	07/07	07/08	07/09
Experiments/ Procedures	Make TB Broth Calculating preparation for malachite green	Machinery research (Preparing lysis protocol, researching in-lab tank sonicator) Innoculating, streaked	Transformation

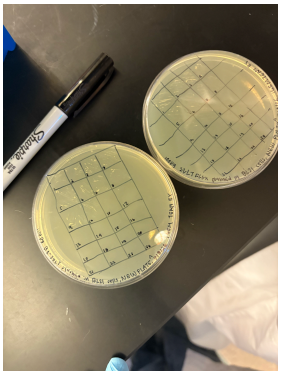
	Researching community autoclave usage Transformation Protocol Outline	plates for colony PCR OD600 Reading	
Key Results	Success	Success	Success

Date: (07/07)

Experiment	Notes	Results	Explanation
TB Broth	(400mL Nanopure water, 2 mL glycerol, 36.8g TB)	Successful	Made TB Broth

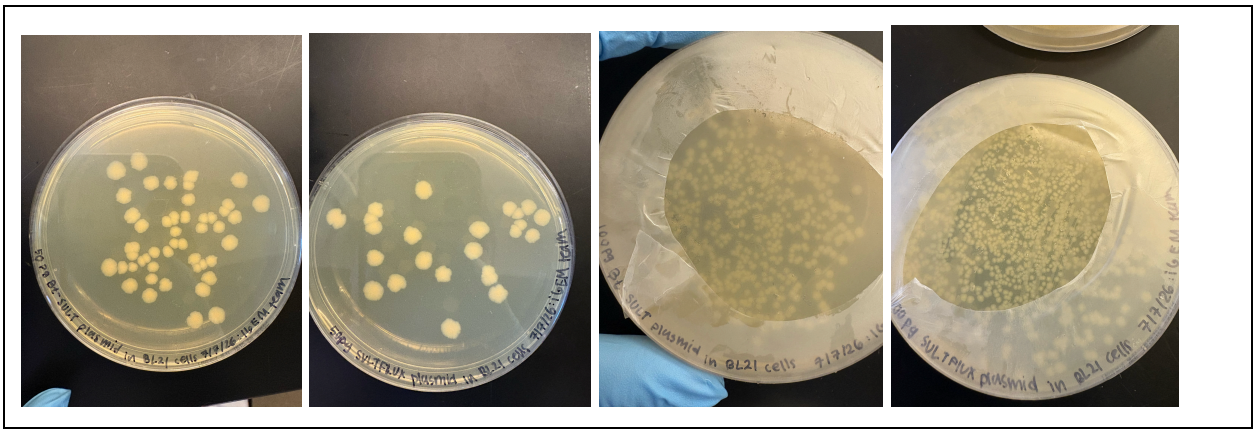
Date: (07/08)

Experiment	Notes	Results	Explanation
Innoculating	(50uL Culture + 5 uL kanamycin)	Success	Colonies appeared



Date: (07/09)

Experiment	Notes	Results	Explanation
Transformation	SF plasmid	Successful colonies	Colonies appeared (1st round of DBTL)



Week 4: (07/13-07/17)

Project Date	**07/13**	**07/14**	**07/15**	**07/16**
Experiments /Procedures	IPTG Stimulation	IPTG Stimulation	BCA	Colony PCR w/ Gel Electrophoresis Protein quantification of SulFlux 1.0 and BtSULT using the NanoDrop - 1:75 had a higher protein conc. (15.0<) compared to 1:100 (2.0>) according to the A280 values on the nanodrop. 1:75 yielded too high of a concentration due to lab constraint times leading to overgrowth. Our team decided to continue with 1:100 so its growth can be more predictable and manipulated.
Key Results	(Relatively) Successful	(Relatively) Successful	Successful	Successful

Date: (07/13)

Experiment	Notes	Results	Explanation
Spiking with IPTG	1:100 (S1), 1:75 (S2), 1:125 (S3) Concentration cultures Auto-cleaving materials Preparing shaker (ordering and measuring lab extensions for beaker/flask specific holders) - OD600 (1.30 hr Check, 2 hr check, 3.30 hr check)	Successful	Increased concentration (barely), evident because of OD600 reading (met ideal standard of 0.4-0.8)

Date: (07/14)

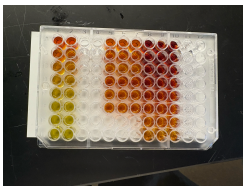
Experiment	Notes	Results	Explanation
IPTG Stimulation	Flasks: 1.) Ctrl (0mM IPTG) 2.) 0.3 mM IPTG 3.) 0.7mM IPTG (Did a ctrl, 1:75, 1:100 dilution)	Successful	Used to conduct a BCA, concentrations increased

Date: (07/15)

Experiment	Notes	Results	Explanation
BCA ASSAY + Data Analysis	(w/ Bpr Lysis, Creation of CBLB Lysis Buffer) | CBLB = B-PER + Lysozyme	Successful	Overall, the concentrations increased overtime. However, the

	+ DNAse		samples are *too concentrated and need to be aliquoted.

Results:



First run

	1	2	3	4	5	6	7	8	9	10	11	12	
A	2.65	2.65	0.03	0.03	2.14	2.22	2.18	*.***	*.***	*.***	0.06	0.06	PBS
B	1.87	1.84	0.05	0.04	2.43	2.22	2.34	*.***	*.***	*.***	0.04	0.04	CBLB
C	1.30	1.30	0.04	0.04	1.91	1.70	1.75	*.***	*.***	*.***	0.04	0.04	
D	1.07	1.07	0.04	0.04	2.96	2.92	3.01	*.***	*.***	*.***	0.04	0.04	
E	0.92	0.92	0.04	0.04	2.03	2.03	2.05	3.71	3.64	3.75	0.04	0.04	
F	0.87	0.85	0.04	0.04	2.11	2.08	2.16	2.56	2.52	2.60	0.04	0.04	
G	0.81	0.81	0.04	0.04	0.04	0.04	0.04	1.81	1.82	1.84	0.04	0.04	
H	0.80	0.80	0.04	0.04	0.04	0.04	0.04	1.40	1.38	1.37	0.04	0.04	

Second Run

	1	2	3	4	5	6	7	8	9	10	11	12
A	2.75	2.74	0.03	0.03	2.19	2.27	2.23	*.***	*.***	*.***	0.06	0.06
B	1.91	1.86	0.05	0.04	2.49	2.28	2.41	*.***	*.***	*.***	0.04	0.04
C	1.34	1.34	0.04	0.04	1.95	1.74	1.79	*.***	*.***	*.***	0.04	0.04
D	1.09	1.08	0.04	0.04	3.04	3.01	3.11	*.***	*.***	*.***	0.04	0.04
E	0.93	0.94	0.04	0.04	2.07	2.08	2.10	3.69	3.61	3.75	0.04	0.04
F	0.88	0.86	0.04	0.04	2.17	2.13	2.21	2.65	2.62	2.69	0.05	0.04
G	0.82	0.82	0.04	0.04	0.04	0.05	0.04	1.86	1.88	1.90	0.05	0.04
H	0.81	0.81	0.04	0.04	0.04	0.04	0.04	1.44	1.42	1.40	0.04	0.04

Date: (07/16)

Experiment	Notes	Results	Explanation
Colony PCR w/ Gel Electrophoresis	Wells labeled: A LADDER, B1-7 LADDER, S1-7 B LADDER, B8-B10, b1-b4 LADDER, S8-S10, su1-su4 C LADDER, b5-b7, su5-su7	SF 1-7 Performed well	Bands showing for S1-7, meaning that gene successfully integrated

Week 5: (07/20-07/24)

Project Date	**07/20**	**07/21**	**07/22**	**07/23**
Experiments/Procedures	Overnight inoculation	BCA 1:2 Dilution, SF + Bt-SULT	Weekly Meeting	HIS-TAG SUMO Pulldown TLC Procedure Outline (Finding concentrations and drafting reaction tubes) Weekly Meeting
Key Results	Successful	-	-	-

Week 6: (07/27-07/31) Start of DBTL2, Part 2

Project Date	**07/27**	**07/28**	**07/29**	**07/30**
Experiments/Procedures			BCA (testing w/ His-tag Binding Buffer) (2 BtSULT colonies, 2 SulFlux colonies)	HIS-TAG Pulldown + DNAse
Key Results	-	-	Successful	Successful

August

Week 7: (08/03-08/07)

Project Date	**08/03**	**08/04**	**08/05**	**08/06**

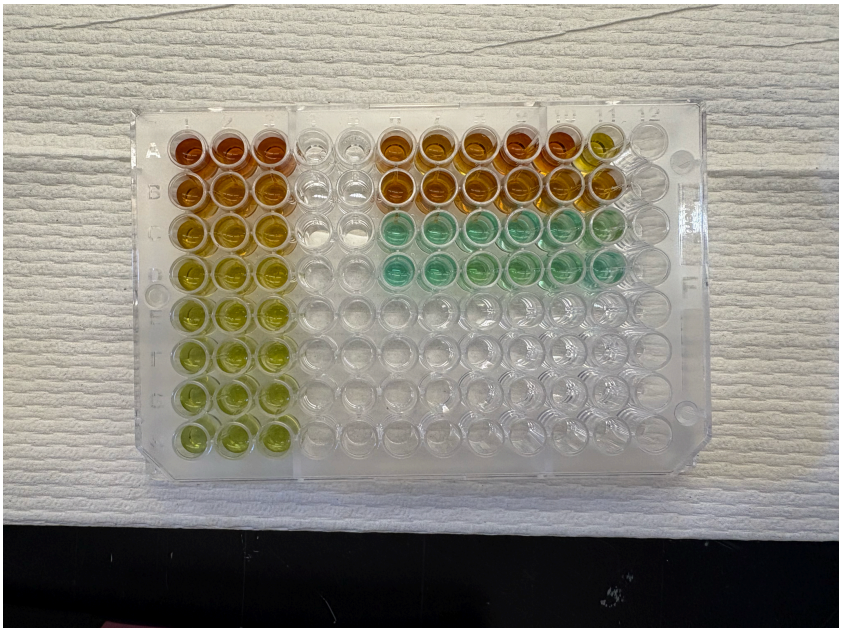
Experiments/Procedures	BCA ASSAY	BCA ASSAY (2) Protein Extraction HIS-TAG Presentation Prep	Weekly Meeting	HIS-TAG, BCA ASSAY
Key Results	Negative	BCA ASSAY I	-	

Date: 08/03

Experiment	Notes	Results	Explanation
BCA ASSAY	Moving forward with His-Tag pulldown procedure on DBTL1 Round1 protein isolation	BCA data analysis showed relatively low conc.	Possible Sonication procedure failed..?

Date: 08/04

Experiment	Notes	Results	Explanation
BCA ASSAY	Used NOT homogenous CBLB (lysozyme and SDS issue)—> from now on, we are going to add the lysozyme and DNase the day of using CBLB	BCA data analysis showed increasing protein concentration	



Date: (08/06) Present: Hannah Bae, Kavi Shrivastava, Dana Tran, Noah Chu, Lakshaya Gupta

Experiment	Notes	Results	Explanation
BCA ASSAY (his-tagged proteins)	-	Successful?	Protein conc. Increasing over time (good), however, it means cells are overcrowding and reaching over the 0.4-0.8 ideal figure. Need to aliquot and lower concentration.

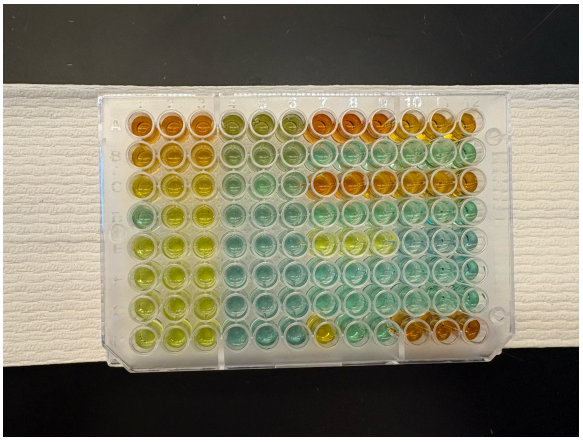


TABLE FOR PROTEIN IN ELUTION	
Concentration(ug/mL)	Avg Standards (nm)
2000	0.845
1000	0.584
500	0.456
250	0.370
125	0.328
62.5	0.328
31.25	0.324
15.625	0.304
7.8125	0.300

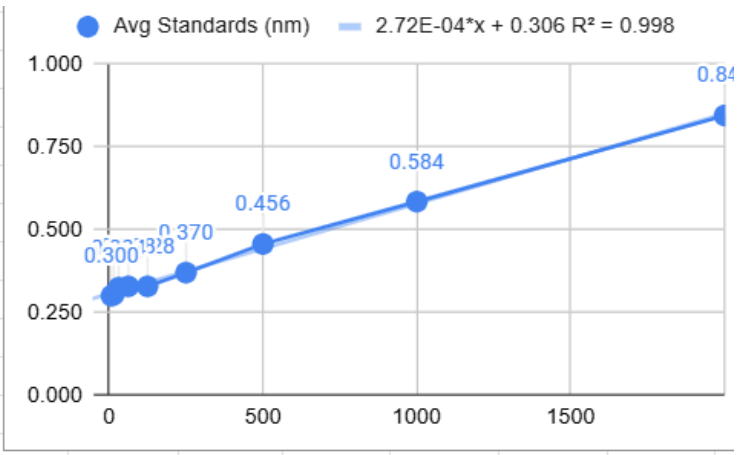
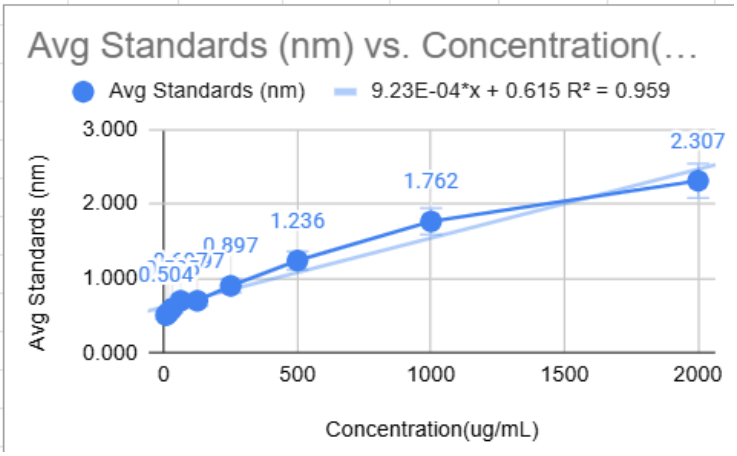


TABLE FOR PROTEIN IN BBB	
Concentration(ug/mL)	Avg Standards (nm)
2000	2.307
1000	1.762
500	1.236
250	0.897
125	0.697
62.5	0.697
31.25	0.595
15.625	0.531
7.8125	0.504



Week 8: (08/10-08/14) No data; no experiments run this week.

Project Date	-	-	-	-	-
Experiments /Procedures	-	-	-	-	-
Key Results	-	-	-	-	-

Week 9: (08/17-08/21) No data; no experiments run this week.

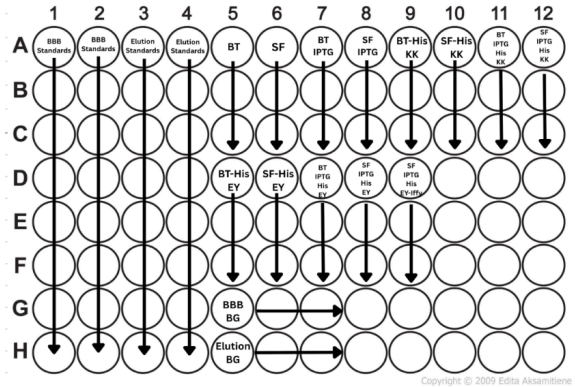
Project Date	-	-	-	-	-
Experiments /Procedures	-	-	-	-	-
Key Results	-	-	-	-	-

Week 10: (08/24-08/28)

Project Date	08/24	08/25	08/26	08/27	08/28
Experiments /Procedures	-	BCA ASSAY PREP	BCA ASSAY	Make TB Broth	-
Key Results	-	-	Low conc.	-	-

Date: 08/26 Present: Dana Tran, Kavi Shrivastava

Experiment	Notes	Results	Explanation
BCA ASSAY	-	Pretty negative/low	Intended range is 0.4-0.8, samples average .2-.3 when blanks are taken out



	1	2	3	4	5	6	7	8	9	10	11	12
A	1.474	0.079	-0.546	-0.581	0.580	0.645	-0.273	0.498	-1.132	-1.128	-1.134	-1.122
B	0.361	0.259	-0.816	-0.832	0.236	-0.683	0.202	0.692	-1.120	-1.126	-1.131	-1.125
C	-0.184	-0.248	-0.962	-0.942	0.303	0.566	-0.255	0.239	-1.115	-1.150	-1.160	-1.144
D	-0.545	-0.565	-1.030	-1.047	-1.100	-1.112	-1.102	-1.109	-1.108	-1.458	-1.462	-1.461
E	-0.743	-0.768	-1.070	-1.090	-1.111	-1.117	-1.107	-1.112	-1.110	-1.461	-1.470	-1.471
F	-0.854	-0.861	-1.082	-1.086	-1.121	-1.110	-1.119	-1.125	-1.109	-1.437	-1.460	-1.442
G	-0.918	-0.925	-1.112	-1.097	-1.450	-1.449	-1.457	-1.454	-1.452	-1.459	-1.453	0.492
H	-0.943	-0.954	-1.106	-1.111	-1.379	-1.425	-1.421	-1.460	-1.466	-1.468	-1.468	0.126

Date: 08/27 Present: Toni Densmore

Experiment	Notes	Results	Explanation
Making TB Broth	500mL TL 47g/1L nanopure water 4mL glycerol Autoclaved @121C for 15 min	Success	Made TB Broth

September

Week 11: (08/31-09/04)

Project Date	**08/31**	**09/01**	**09/02**	**09/03**	**09/04**
Experiments /Procedures	-	Transformation Prep	Transformation	Transformation	-
Key Results	-	-	Failed	Failed	-

Date: (09/02) Present: Dana Tran, Kavi Shrivastava

Experiment	Notes	Results	Explanation
Transformation	**SF2: F1475 **SF3: F1475_E144D SF2 @ 100pg/ul	Failed, no colonies	Too low of concentration?

Date: (09/03) Present: Dana Tran, Kavi Shrivastava

Experiment	Notes	Results	Explanation
Transformation	SF2 and SF3 @ 100pg/ul and 1 ng	Failed, no colonies	Too low of concentration?

Week 12: (09/07-09/11)

Project Date	**09/07**	**09/08**	**09/09**	**09/10**	**09/11**
Experiments /Procedures	-	-	Transformation	Transformation of DH5alpha (control)	Transformation (continued)
Key Results	-	-	Failed	Failed	Failed

Date: (09/09) Present: Kavi Shrivastava, Dana Tran, Toni Densmore

Experiment	Notes	Results	Explanation
Transformation	SF3, SF2, Bt-SULT (50pg/ul aliquots of DNA each) Old (ctrl): BTH52KK New: SFH52KK and BTH53KK	BTH52KK: Failed SFH52KK: Failed BTH53KK: Failed	Human error? Suspect that plasmids sent are malfunctioning.

	SF2/SF3 @ 100pg/uL		

Date: (09/10-09/11) Present: Dana Tran, Kavi Shrivastava

Experiment	Notes	Results	Explanation
Transformation Preparation	Multiple failed transformations; used PUC19 as ctrl to test if there were issues with Bt-SULT and SF plasmids DH5alpha + PUC19 DH5alpha + SF2 DH5alpha + SF3	Failed. No colonies formed.	May be human error, concentration of plasmid too low, so didn't transform? Will use PUC19 as a control to test issues with our plasmid.

Week 13: (09/14-09/18)

Project Date	**09/14**	**09/15**	**09/16**	**09/17**	**09/18**
Experiments /Procedures	-	-	Transformation	-	-
Key Results	-	-	Failed	-	-

Date: (09/16) Present: Toni Densmore

Experiment	Notes	Results	Explanation
Transformation	PUC19 CTRL	Failed.	-

Week 14: (09/21-09/25)

Project Date	**09/21**	**09/22**	**09/23**	**09/24**	**09/25**
Experiments /Procedures	-	Transformation	BCA ASSAY	-	-
Key Results	-	PUC19 CTRL, successful	Mostly negative	-	-

Date: 09/23 Present: Kavi Shrivastava, Dana Tran

Experiment	Notes	Results	Explanation

BCA ASSAY	Pooled all SF and BT samples (His-tagged) → BCA for final conc.	Very low, mostly negative	Our samples did not survive

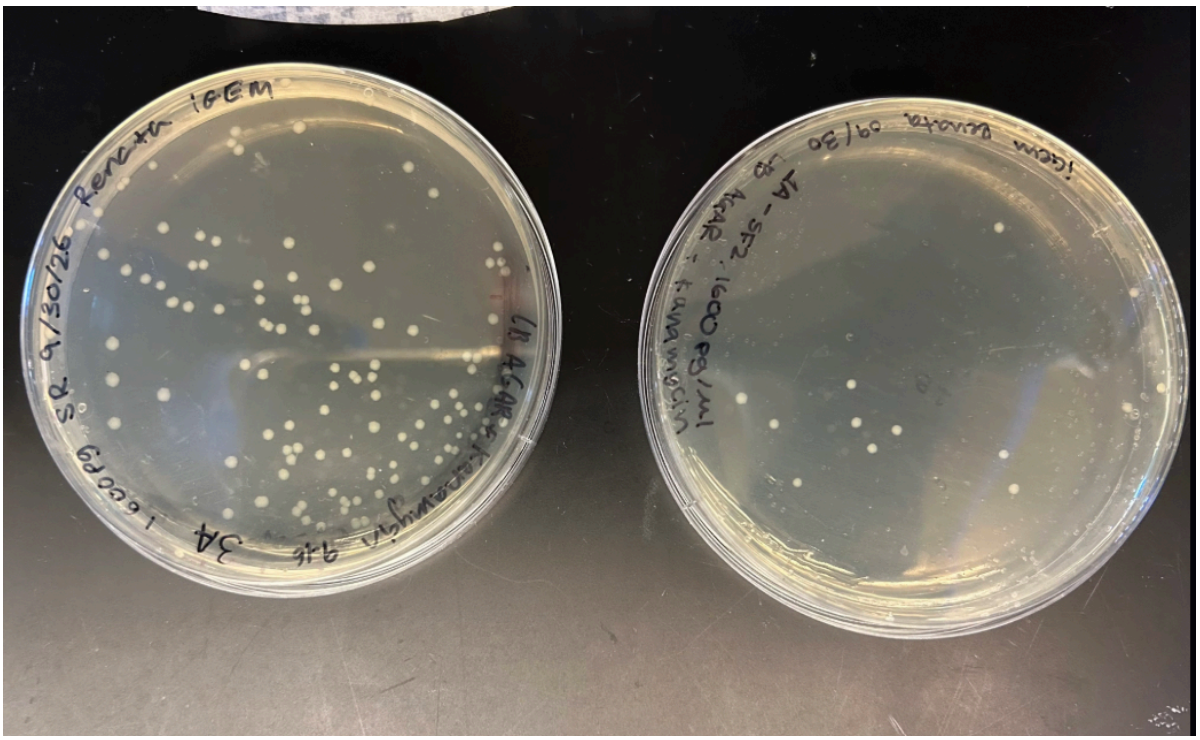
October

Week 15: (09/28-10/03)

Project Date	**09/29**	**09/30**	**09/31**	**10/01**	**10/03**
Experiments/Procedures	-	Transformation	-	Colony Restreak	Colony PCR + Visualization
Key Results	-	Success	-	-	Bands for SF2 and SF3 Neg: 1AA, 2AA Pos: 1AB, 2AB, 3AA, 3AB

Date: 09/30-10/01 Present: Kavi Shrivastava, Dana Tran, Toni Densmore

Experiment	Notes	Results	Explanation
Transformation	1A: SF2 @ 1600pg/ul 2A: SF3 @ 1600pg/ul 3A: BtS @ 1600pg/ul 1B: SF2 @ 800pg/ul 2B: SF3 @ 800pg/ul 3B: BtS @ 800pg/ul	Successful 1A: 13 colonies 1B: 19 colonies 2A: 13 colonies 3A: 80-100 colonies 3B: 180-200 colonies	Spun down sample during transformation, aliquoted heavily condensed sample into bacterial tubes to shake and grow



Date: 10/03 Present: Kavi Shrivastava, Dana Tran, Toni Densmore

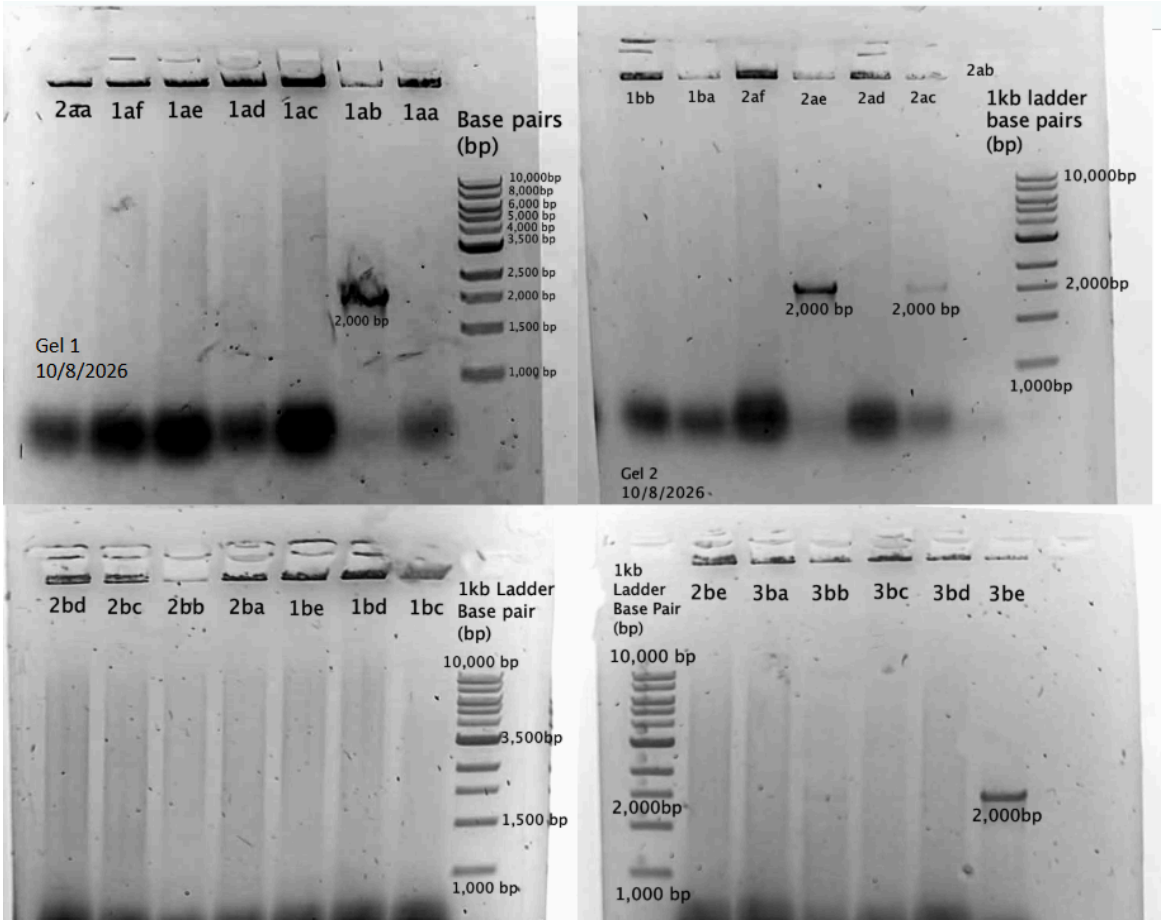
Experiment	Notes	Results	Explanation
Colony PCR	-	1aa (SF2 @ 1600pg/ul), 2aa (SF3 @ 1600pg/ul)	Most of our samples did not show bands (negative). The only two positive results were 1aa and 1aa.
Colony PCR image:			

Week 16: (10/05-10/10)

Project Date	**10/05**	**10/06**	**10/07**	**10/08**	**10/10**
Experiments /Procedures	-	-	Restreak Colony PCR Prep	COLONY PCR TB Broth	IPTG Spike
Key Results	-	-	Success	Success	Success

Date: (10/08) Present: Toni Densmore, Kavi Shrivastava, Dana Tran

Experiment	Notes	Results	Explanation

Colony PCR	Ran 4 gels for: 1aa-af, 2aa-af	Mild banding	Some of our samples showed positive, others negative.
Colony PCR images: 			
TB Broth	-	-	-

Date: (10/10) Present: Kavi Shrivastava

Experiment	Notes	Results	Explanation
IPTG SPIKE	Spiked 1AB, 2AE, 3BE OD600 Reading (2hrs) 1AB: 0.7 2AE: 0.492 3BE: 0.717 Samples split into two	Spiked everything with IPTG	Increased the concentration of our samples

	flasks with IPTG, one with Milk		

Week 17: (10/11-10/17)

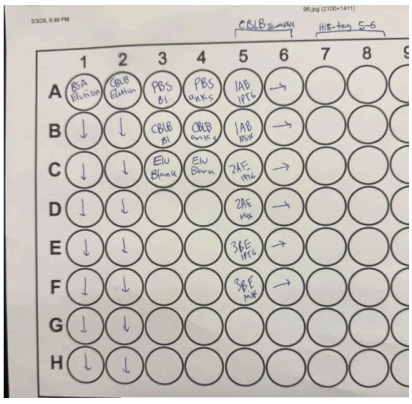
Project Date	**10/11**	**10/13**	**10/14**	**10/15**	**10/17**
Experiments /Procedures	Protein Extraction	SDS PAGE PREP BCA ASSAY	-	Calculations + Prep for SDS-PAGE and TLC	SDS-PAGE TLC
Key Results	Success	Success	-	-	SDS PAGE Failed TLC Failed

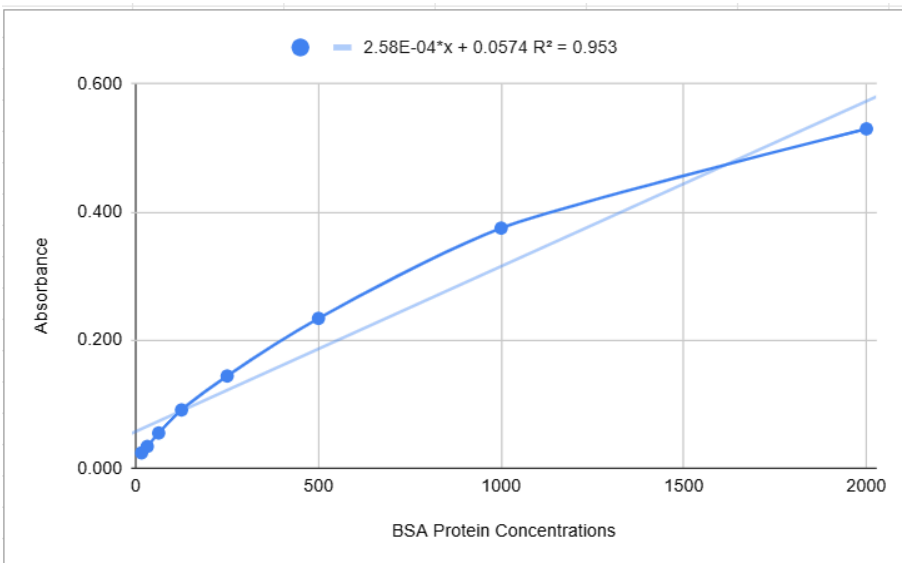
Date: (10/11) Present: Kavi Shrivastava, Toni Densmore

Experiment	Notes	Results	Explanation
Protein Extraction	CBLB made fresh 10-12 BSA Elution marks A-H	Successful	-

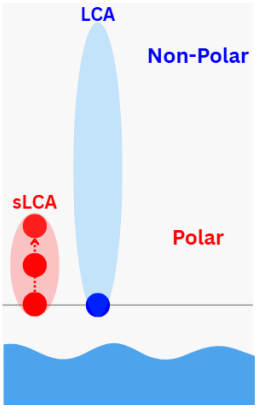
Date: (10/13) Present: Toni Densmore, Hannah Bae

Experiment	Notes/Images: Overall BSA ASSAY conc. increasing
BCA ASSAY	
Well Planner	



OD600 Reading	

Date: 10/17 Present: Toni Densmore, Kavi Shrivastava, Dana Tran

Experiment	Notes	Results	Explanation
SDS PAGE	-	No Bands (Failed)	Possible Causes: Overloaded wells, did not refrigerate running buffer
SDS PAGE IMAGES:			
Thin Layer Chromatography (TLC)	Practice run for official TLC on 10/19 	Inconclusive (Failed)	Gypsum deposits on TLC plates; gypsum serves as a binder and caused plates to no longer bind Possibility: did not dry plates well enough before use.

Week 18: (10/19-10/24)

Project Date	**10/19**	**-**	**-**	**-**	**-**
Experiments	SDS Page	-	-	-	-

/Procedures	TLC				
Key Results	SDS Page Success, TLC failed	-	-	-	-

Date: 10/19 Present: Kavi Shrivastava, Dana Tran, Toni Densmore

Experiment	Notes	Results	Explanation
SDS-PAGE	Had to use a stained ladder, visualized with Odyssey CLx. Refrigerated running buffer prior to use (better conduction) Empty wells were filled with LDS (loading buffer) as placeholders 50ul protein ladder (stained), 30ul per protein Ran for 1.5 hr, 100v, 25W	Bands (Success)	
Thin Layer Chromatography	Experimentation with the possibility of using acetic acid and UV fluorescence to quantify LCA reactions.	Inconclusive (Failed)	No standard visualization acid available